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RESEARCH ARTICLE

Outlier Detection of Photovoltaic Power Generation System Data Based on GMM-IF Algorithm

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ABSTRACT The Gaussian Mixture Model (GMM) is widely used in anomaly detection due to its flexibility in handling complex data distributions and its soft classification mechanism. However, its modeling process relies on the assumption that data follows a Gaussian distribution. In contrast, time-series data from photovoltaic (PV) systems often exhibit high nonlinearity and non-stationarity, making it difficult to represent with a single Gaussian distribution, which affects detection performance. To address this issue and improve detection accuracy, this paper proposes a segmented GMM-IF outlier detection algorithm. In this method, the decision tree segmentation modeling strategy is used to structure the data to construct a local Gaussian-like distribution dataset to overcome the prior assumptions. Additionally, it integrates an Isolation Forest (IF) as the decision-making mechanism for detailed detection, further enhancing detection accuracy. Under different operating conditions and actual photovoltaic array datasets, comparative experiments were conducted using different evaluation metrics and algorithms to verify the effectiveness and superiority of the GMM-IF algorithm. Experimental results show that compared with the single model of GMM and IF, the GMM-IF algorithm has an average improvement of 3.8% in accuracy and 9.1% in F-value. Compared with other comparison algorithms, GMM-IF has obvious advantages, especially in terms of recall.

INDEX TERMS Photovoltaic power generation system, outlier detection, GMM algorithm, decision tree, IF.

I. INTRODUCTION

In recent years, environmental pollution is serious and energy crisis has arrived. The development and use of new energy have been highly concerned [1], [2]. The solar energy has the characteristics of large reserves, wide distribution and high efficiency. So it is widely used in the new energy field [3]. According to the International Agency (IEA) energy report, the global PV total has reached 760 GW by the end of 2020.

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It is expected to reach 1,721 GW by 2030 and to further increase to 4,670 GW by 2050 [4]. With the development of the PV industry and the increase of the total installed capacity of PV globally, it is particularly important to ensure the normal operation and maintenance of PV systems.

Due to the complex external situation of the PV array operation, the output power is susceptible to the influence of the environment such as temperature and irradiance, coupled with the abnormal operation of the PV equipment, which will lead to a large number of outliers in the output power. A large number of outliers will affect the realization of system

performance analysis, fault diagnosis, state evaluation, power prediction and other functions [5]. Therefore, the issue of outlier detection in photovoltaic power generation systems urgently needs to be addressed.

Outliers are data points that are significantly different from other data points or that do not mimic the expected typical behavior of other data points [6]. Outlier detection is an effective data analysis method for identifying abnormal data in a data set. The main methods used in data-driven outlier detection include: distance-based algorithms [7], [8], density-based algorithms [9], [10], statistic-based algorithms [11], [12], cluster-based algorithms [13], [14] and algorithms based on deep learning. Distance-based and density-based methods typically rely on data labels. Deep learning-based methods, although possessing strong feature extraction capabilities, are limited by the amount of training data and computational resources. Clustering-based methods face significant uncertainty in the number of clusters. Therefore, the GMM algorithm, a typical representative of statistical-based methods, is used as the core for detection. Improvements are made to overcome the prior assumptions and enhance detection accuracy, thus leading to the proposed GMM-IF algorithm.

II. RELATED WORK

In recent years, deep learning-based methods have achieved significant results in the application of large-scale datasets. Harrou [15] combined the feature extraction capability of variational autoencoders with a variety of semi-supervised outlier detection algorithms to enhance the fault detection of grid-connected photovoltaic systems. This method is limited by variational autoencoders, which can easily lead to information loss during the coding process, thus reducing the detection accuracy. He et al. [16] combined Bayesian neural network and Monte Carlo estimation to propose a new probabilistic outlier detection. This method improves the reliability of time series anomaly detection of photovoltaic system. However, it does not focus on the detection accuracy of time series.

In the distance-based algorithm, Zou et al. [17] proposed a method based on KNN combined with local outlier factors. The continuity and integrity of time series are fully considered, and K value is determined by elbow method. But detection results depend on data labels. Li et al. [7] used kernel density estimation and locally accessible distance to locate abnormal regions. Outliers are determined according to the proposed density lifting distance as a measure. This method overcomes the dependence on data labels. However, it does not have good performance ability under the condition of high nonlinear degree and short time fluctuation. Harrou et al. [18] uses a single-diode model to describe the normal state of the photovoltaic system, and introduces a one-class SVM to describe the residuals to detect anomalies. To a certain extent, this method overcomes the influence of nonlinearity on the detection progress, but it has shortcomings in distinguishing between normal fluctuations and abnormal faults. In the density-based outlier detection,

Ding et al. [19] combined the local anomaly factor outlier detection algorithm and Grubbs criterion to detect the current data of the PV array simulation model and then carry out fault diagnosis. The method depends on the selection of hyperparameters. Different hyperparameters have a great influence on the detection results. Sheng et al. [20] proposed a weighted Gaussian process regression method to assign lower weights to outliers with higher anomalous potential values. In addition, a density-based local outlier detection method is introduced to compensate for the deterioration of high dimensional distance and complete the short-term photovoltaic power prediction. This method is also restricted by the selection of hyperparameters. Taghezouit et al. [21] uses kernel density estimation to set the residual threshold for the empirical model with double exponential weighted average, which gets rid of the constraints of some hyperparameters, but is limited by the adaptability of model assumptions. Outlier detection methods based on clustering are easily constrained by data labels. Ying et al. [22] used mean shift to cluster data, then used MKNN algorithm to automatically label the data, and finally used average weighting technology to detect outliers. The shortcoming of insufficient labeled sample size in the data is overcome. However, in the process of use, the number of clusters is usually uncertain. If the number of clusters is set to 2, the detection accuracy is not high due to the influence of noise. This is also a common disadvantage of cluster-based outlier detection. However, the statistic-based approach has obvious advantages in fitting data distribution, interpretability and robustness.

Gaussian mixture model is a typical outlier detection method based on statistics. Since the Gaussian mixture model was proposed, it has been favored by researchers because of its ability to fit various data efficiently. Bishop et al. [23] use the mixed density network to improve GMM. The ability of GMM algorithm for nonlinear fitting has been enhanced, enabling end-to-end learning and multimodal distribution modeling. Since then, GMM algorithm has been widely concerned. Qiao et al. [24] firstly clusters two-dimensional data with k-means to obtain the number of clusters, and then uses GMM to perform data regression. But the method is easily affected by data density. When the density is close or the distance between outliers and normal data is small, the detection accuracy is low. Liu et al. [25] uses high-order Markov chain and Pearson correlation coefficient to obtain the state transition probability matrix to construct GMM. Liang et al. [26] combines the invisible Markov chain with GMM to build a hybrid model. It is similar to the method proposed by Liu. The model only considers the influence of the first few moments on the data. Fang et al. [27] proposes a data cleaning algorithm based on kernel density estimation to eliminate the prior assumption that all parts of GMM need to conform to Gaussian distribution. On the basis of using two-dimensional singular value decomposition to preserve the correlation and dependence between time series. Ge et al. [28] uses sparse autoencoder and GMM to build models to improve the generalization ability of models. But

this method is not suitable for time series with a single variable. In the detection of outliers in time series, the above methods can not satisfy the premise of GMM well and cannot detect outliers that are closer to normal data.

In view of the shortcomings of the above GMM-based algorithm, an anomaly detection method called GMM-IF was proposed, and it was applied to the detection of abnormal data of photovoltaic systems. The GMM-IF algorithm consists of three stages. In the first stage, the photovoltaic output power data is segmented by decision tree algorithm. The decision tree can automatically identify regions with similar distributions based on the characteristics of the data, thereby dividing the dataset into multiple subsets. Each subset better fitting a Gaussian distribution. In the second stage, the segmented data subsets are applied to the GMM model. After modeling each subset, the corresponding probability distributions are generated, and potential outliers in the data are detected based on these probability distributions. The main reason for using decision tree segmentation is to overcome the prior assumptions of the GMM model and improve the detection of outliers (i.e., edge outliers) distributed near the main output power. Therefore, the obtained detection method combines the flexibility of the decision tree with the objectivity of the GMM algorithm in finding photovoltaic anomalies. The method of segmenting photovoltaic data using decision trees effectively alleviates the dependency of the traditional GMM algorithm on the Gaussian distribution assumption. However, the segmented data still exhibits some deviation, which leads to certain limitations in the performance of anomaly detection. Therefore, the isolated forest algorithm was introduced in the third stage to classify the original data. It is used as a decision-making and judgment mechanism, and works in conjunction with the outlier detection algorithm to jointly complete the outlier detection. Compared to traditional methods, the GMM-IF algorithm not only overcomes the GMM's requirement for data distribution, but also enhances the robustness of anomaly detection in complex photovoltaic data through the introduction of Isolation Forest. Especially when facing highly nonlinear data, it still performs well, demonstrating stronger adaptability and higher detection accuracy.

The following arrangements are made in the following sections of this paper: In Chapter 3, the principle of the proposed segmented GMM-IF algorithm is introduced. In Chapter 4, the experiment and analysis of simulation data and real data are carried out. In Chapter 5, the conclusion is summarized.

III. THE PRINCIPLE OF SEGMENTED GMM-IF ALGORITHM

A. BASAL PRINCIPLE

1) GAUSSIAN MIXTURE MODEL

The Gaussian mixture model consists of Gaussian distributions in multiple high-dimensional Spaces, each of which has a different proportion in the whole. The Gaussian mixture

model is expressed as:

$$P(x) = \sum_{k=1}^K p_k N(x|\mu_k, \Sigma_k) \quad (1)$$

where, $\sum_{k=1}^K p_k = 1$, p_k represents the weight of the k -th Gaussian distribution. $N(x|\mu_k, \Sigma_k)$ represents the probability density function of the k -th Gaussian distribution. In a higher dimensional space, the Gaussian distribution is defined as:

$$P(x|\mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \quad (2)$$

where, d represents the dimensions of the Gaussian distribution, μ represents the mean, Σ represents the variance.

2) ISOLATION FOREST

Isolated forest is composed of multiple isolated trees. Isolation Tree is a binary tree. Each step of segmentation divides the data set into two parts. When segmentation is carried out on each node, the feature value and segmentation value are selected until one of the following conditions is met:

- (1). The tree model reaches its maximum depth.
- (2). There is only one sample on this node.
- (3). The eigenvalues of all samples on this node are the same.

B. THE PRINCIPLE OF SEGMENTED GMM-IF ALGORITHM

In this part, the segmentation GMM-IF algorithm is proposed to detect outliers and identify observations that are significantly different from other data points (e.g. non-stationary noise) in the data, so as to provide clean and reliable data for later work as much as possible. This observation does not include anomaly class data (e.g., stationary noise) so as not to adversely affect related work (e.g., fault diagnosis, feature extraction). In the process of using GMM algorithm, the number of clusters is not known in advance. And to improve adaptability, data segmentation is carried out in the first stage. In the second stage, Gaussian mixture model is used to detect outlier points on the basis of segmentation. In the third stage, the isolated forest is used as the judgment mechanism to detect outliers on the whole data and cooperate with the results of the second stage to complete the outlier detection. The block diagram of GMM-IF outlier detection is shown in Figure 1.

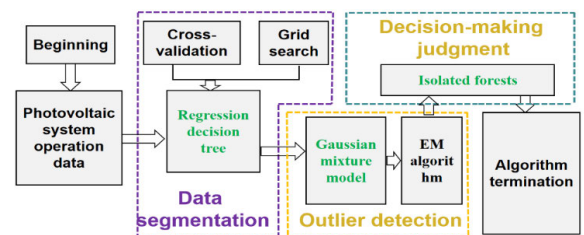


FIGURE 1. Block diagram of GMM-IF.

1) ROUGH OUTLIER DETECTION ALGORITHM BASED ON SEGMENTED GMM-IF

Decision tree is a tree-like structure, which has the advantages of easy integration and no need to preprocess data. Each of its internal nodes represents a judgment, each branch represents the result of a judgment, and the last node of each branch represents a final result. Decision tree is essentially a tree structure composed of multiple judgment nodes. This model can be used for both prediction and regression. Decision trees for regression are used to divide the data into several pieces.

The core problem of decision tree regression is to determine the output values of partition points and leaf nodes. In the data set, each data has n features and each feature has $s_i (i \in (1, n))$ values. After traversing all features and the corresponding value of all features, the space is divided until the mean squared error is minimized. At this time, the corresponding value of the feature and feature is the value of the internal node of the decision tree. To wit:

$$\min_{j,s} [\min_{c_1} \sum_{x_i \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j,s)} (y_i - c_2)^2] \quad (3)$$

where, x_i represents the input eigenvector of sample i . j is the feature index. s is the segmentation value corresponding to feature j . $R(j, s)$ is the data set after this segmentation and before the next segmentation. c_1 and c_2 represent the predicted values of the target variable y on the segmented data set.

Using the above formula, the data set is segmented with the selected (j, s) , and the corresponding output value is obtained:

$$R_1(j, s) = \{x | x^{(j)} \leq s\}, R_2(j, s) = \{x | x^{(j)} > s\} \quad (4)$$

$$c_m = \frac{1}{N_m} \sum_{x_i \in R_m(j,s)} y_i \quad m \in 1, 2 \quad (5)$$

By repeating the above (3) and (4), the space can be divided into M regions, denoted as: $A = \{A_1, A_2, \dots, A_M\}$. The decision tree can be generated:

$$f(x) = \sum_{m=1}^M \hat{c}_m I(x \in A_m) \quad (6)$$

where I is the indicator function,

$$I = \begin{cases} 1 & \text{if } (x \in A_m) \\ 0 & \text{if } (x \notin A_m) \end{cases}$$

To avoid overfitting of the model and reduce the effect of outlier detection, grid search is used to prune the decision tree and the k-fold cross-validation was used to improve the generalization ability of the model and avoid overfitting.

Grid search is a method to optimize hyperparameters. The first step is to confirm the hyperparameters to be tuned and their ranges. And it is traversed by exhaustive method. Then the hyperparameters are put into the model to be trained and the cross-validation is used to select the optimal parameter. The selected hyperparameters are the maximum depth of the

decision tree and the minimum number of samples of leaf nodes.

Cross-validation is a technology that repeatedly uses data and constantly selects different data for training, testing and evaluation. The flow chart of cross-validation is shown in Figure 2. Assuming that the data includes N samples, divide the data into k equal parts, denoted as: $D = \{D_1, D_2, \dots, D_k\}$. Each sample contains N/k samples, A total of k loops are traversed. In each cycle i , D_i is taken as the test set and the union of the remaining subsets is taken as the training set that is denoted as: $D_{part} = D - D_i$. MSE is used as the evaluation index. The average of MSE is taken to complete the model evaluation.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7)$$

where, y_i is the actual value. $\hat{y}_i$ is the predicted value. n is the number of samples.

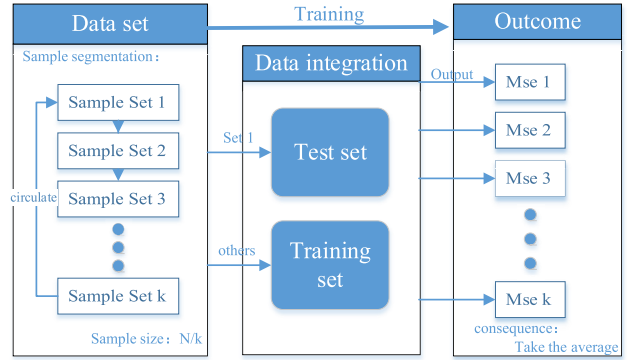


FIGURE 2. Flow chart of isolated forests.

The data set is divided into several segments. Then GMM algorithm is used to detect outliers on the basis of segmentation. Assume that the operating power of the PV module conforms to the Gaussian mixture model in a given segment. The complete data in each segment is represented as (X, Z) . There are N groups of data. $X = (x_1, x_2, \dots, x_N)$ is the observation data. x_k is the observation value that is the output power of the PV panel at different times. $Z = (z_1, z_2, \dots, z_N)$ cannot be obtained by observation. z_k is a hidden variable that indicating the corresponding sample x belongs to which Gaussian distribution. Assume parameters $\theta = \{p_1, p_2, \dots, p_K, \mu_1, \mu_2, \dots, \mu_K, \Sigma_1, \Sigma_2, \dots, \Sigma_K\}$. After introducing hidden variables, the joint probability distribution of the Gaussian mixture model is:

$$P(x, z) = P(z)P(x|z) = p_z N(x|\mu_z, \Sigma_z) \quad (8)$$

At this time, under the condition of known observation data, the conditional probability distribution of the hidden variable is:

$$P(z|x) = \frac{P(x, z)}{P(x)} = \frac{p_z N(x|\mu_z, \Sigma_z)}{\sum_{k=1}^K p_k N(x|\mu_k, \Sigma_k)} \quad (9)$$

The maximum likelihood estimate $\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} \log P(x)$ should be used to solve p_k , μ_k and Σ_k for of the above equation. However, since there is no analytical solution for $\hat{\theta}_{MLE} = \arg \max_{\theta} \sum_{i=1}^N \log \sum_{k=1}^K p_k N(x_i | \mu_k, \Sigma_k)$, the EM algorithm is used to solve p_k , μ_k and Σ_k .

EM algorithm is a two-step algorithm that iteratively solves hidden variables, namely E Step (Expectation Step) and M step (Maximization Step). The mathematical expression of EM algorithm is:

$$\theta^{(t+1)} = \underset{\theta}{\operatorname{argmax}} E_{Z|X\theta^{(t)}}[\log P(X, Z|\theta)] \quad (10)$$

In step E, let $Q(\theta, \theta^{(t)}) = E_{Z|X\theta^{(t)}}[\log P(X, Z|\theta)]$. According to the results of the previous iteration, it is solved:

$$\begin{aligned} Q(\theta, \theta^{(t)}) &= \sum_{Z_1, Z_2, \dots, Z_N} [\log P(x_1, z_1|\theta) + \dots + \log P(x_N, z_N|\theta)] \\ &\quad \prod_{i=1}^N P(z_i | x_i, \theta^{(t)}) \\ &\quad \sum_{Z_1, Z_2, \dots, Z_N} [\log P(x_1, z_1|\theta)] \prod_{i=1}^N P(z_i | x_i, \theta^{(t)}) = \sum_{Z_1} [\log P(x_1, z_1|\theta)] \end{aligned} \quad (11)$$

$P(z|x, \theta^{(t)})$. So :

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^N \sum_{Z_i} [\log P(x_i, z_i|\theta)] P(z_i | x_i, \theta^{(t)}) \quad (12)$$

where the posterior probability is equal to the conditional probability distribution of the hidden variable in the Gaussian mixture model, that is:

$$P(z|x) = \frac{p_z N(\mu_z, \Sigma_z)}{\sum_{k=1}^K p_k N(x|\mu_k, \Sigma_k)} = P(z_i | x_i, \theta^{(t)}).$$

is equal to the joint probability distribution of the Gaussian mixture model, thus:

$$\begin{aligned} Q(\theta, \theta^{(t)}) &= \sum_{i=1}^N \sum_{z_i} [\log p_{z_i} N(x_i | \mu_{z_i}, \Sigma_{z_i})] \\ &\quad \frac{p_{z_i} N(x_i | \mu_{z_i}^{(t)}, \Sigma_{z_i}^{(t)})}{\sum_{k=1}^K p_k^{(t)} N(x | \mu_k^{(t)}, \Sigma_k^{(t)})} \end{aligned} \quad (13)$$

In step M, the result of step E is used to solve the maximum likelihood function. Then update p_k , μ_k , and Σ_k . Bringing (13) into (10) and using the Lagrange multiplier method to solve, we get:

$$p_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N P(z_i = C_k | x_i, \theta^{(t)}) \quad (14)$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^N x_i P(z_i = C_k | x_i, \theta^{(t)})}{\sum_{i=1}^N P(z_i = C_k | x_i, \theta^{(t)})} \quad (15)$$

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^N (x_i - \mu_k)^T (x_i - \mu_k) P(z_i = C_k | x_i, \theta^{(t)})}{\sum_{i=1}^N P(z_i = C_k | x_i, \theta^{(t)})} \quad (16)$$

The Expectation Step and the Maximization Step are carried out alternately until the error is extremely small and the iteration stops to complete the model and data fitting. Then the data classification of outliers and non-outliers is completed by setting thresholds.

2) DECISION AND JUDGMENT MECHANISM OF MULTI-MODEL FUSION

The isolated forest is used to optimize the detection results to improve the detection accuracy of the algorithm after using the Gaussian mixture model to detect outliers on the basis of segmentation. By using the method of multi-model fusion, the data that are jointly detected as outliers are identified as outliers by setting appropriate thresholds.

Isolated forest is an algorithm to detect outliers by describing the degree of alienation. Isolated forests have significant advantages in dealing with simple distribution, noise and outlier detection. By constructing an isolated tree and forming an isolated forest, the nodes that are close to the root node and have no child nodes are identified as outliers. The specific process of building an isolated forest is as follows:

(1). Perform data sampling. Select a part of the samples, the number of which is much smaller than the data set, so as to reduce the complexity of the calculation;

(2). Build an isolated tree using the selected samples. Firstly, a feature is selected randomly. Then a cut value is randomly selected based on the feature selection. The cut value needs to be divided into two parts of the data. The data is then cut into two parts with the cut value of the selected feature. Finally, repeat the above operation for the two parts that are cut until the conditions for stopping the cutting are met.

(3). Isolated trees are formed into isolated forests

The flow diagram for building an isolated forest is shown in Figure 3.

The outlier detection was carried out using the constructed isolated forest and the outlier judgment was carried out based on the path length. First calculate the path length $h(x)$:

$$h(x) = e + c(T.size) \quad (17)$$

where, e represents the number of paths that a certain sample goes through from the root node to the leaf node of an isolated tree, that is, the number of cuts. $T.size$ represents the number of other samples in the same leaf node as this sample. The introducing of $c(T.size)$ can make a greater difference in path

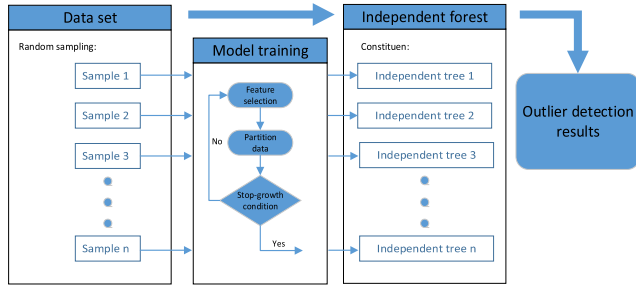


FIGURE 3. Cross validation flow chart.

length between normal and abnormal samples. $c(n)$ is:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \quad (18)$$

where, $H(n-1) = \ln(n-1) + 0.5772156649$, 0.5772156649 is Euler's constant.

The anomaly score $s(x, n)$ is used as the criterion for judging outliers:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (19)$$

where, $E(h(x))$ represents the average path length. The value range of the exponential range is $(-\infty, 0)$. The value range of the whole range is $(0, 1)$. In the IF, the anomaly score of each isolated tree is calculated and the average is taken at last. The higher the anomaly score, the closer it is to the root node, the more likely it is to be an outlier. A simple isolated forest is shown in Figure 4. The normal data points are green. the outliers are red. The darker the color of the blue data points. The darker the color of the blue data points, the larger the outlier, the more likely it is to be an outlier.

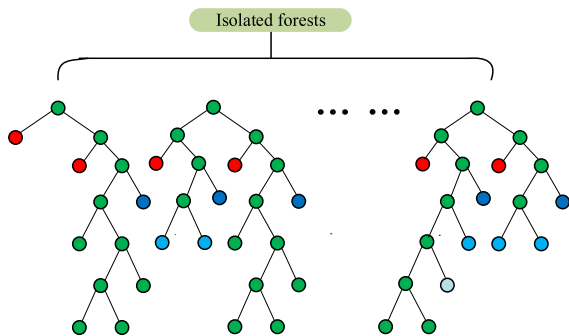


FIGURE 4. A simple isolated forest.

In the process of decision tree training, each partition needs to traverse the dataset. The depth of the tree is about $O(\log n)$. So the time complexity of the decision tree division dataset part is $O(n \log n)$, where n is the number of samples. The GMM time complexity is based on the EM algorithm. Each iteration needs to calculate the probability that all sample data belong to each Gaussian distribution, so its time complexity is $O(knd^2)$. Where k is the number of mixture components and d is the data dimensionality. Assuming that the number of EM iterations is T , the time complexity of GMM is $O(Tknd^2)$.

IF consists of T isolated trees, so the time complexity is $O(m \log n)$. In addition, the time complexity of the fusion judgment part is $O(n)$, which is negligible. Therefore, the time complexity of the GMM-IF algorithm is:

$$O(n \log n + Tknd^2 + m \log n)$$

IV. SIMULATION AND EXPERIMENT

The experimental environment of this experiment is AMD R5 CPU, 4800 MHz 16 GB memory and Windows 11 operating system. Python was used to implement the algorithm, and Matlab was used to build the PV system. The software environment is Python3.7.0 and Matlab2022a.

In this part, the GMM-IF algorithm is applied to simulation data and real data respectively, which proves the feasibility of the proposed algorithm. Compared with IF, GMM, GMM-RF, GMM-NN, SVM and LOF algorithms, the advantages of the proposed algorithms are proved. Among them, both the GMM-RF and GMM-NN algorithms are improved based on the GMM algorithm. The comparative analysis of the relevant algorithms based on GMM is shown in Table 1.

TABLE 1. Comparative analysis table of algorithms based on GMM.

Algorithms	Core Optimization Methods	Algorithm Analysis
GMM	Iterative Optimization of Gaussian Components Based on EM Algorithm	Classical Theoretical Support (Suitable for Simple Distribution Modeling)
GMM-RF	Integrating Random Forest for Feature Screening and Nonlinear Modeling	High-dimensional data screening capability is strong, but low-dimensional data is prone to overfitting.
GMM-NN	Integrating Neural Networks to Enhance Complex Distribution Fitting Capability	Suitable for complex data but requires long training time and depends on large datasets.
GMM-IF	Decision Tree Segmentation and Multi-Model Fusion Judgment	High detection accuracy, especially in low-nonlinearity scenarios.

A. SIMULATION DATA SET CONSTRUCTION

The simulation data set is constructed by means of simulation and manual. The build process is divided into three parts: First, refer to the typical literature and guidelines in the photovoltaic field to determine the classification of photovoltaic system outliers. Then, the PV system was simulated and data collected in 6 s as the base unit. The simulation diagram of PV system Simulink is shown in Figure 5. Finally, according to the different classification and characteristics of outliers, outliers are added to the simulation data set.

1) SOURCE AND CLASSIFICATION OF OUTLIERS

The common types of outliers in PV system data are divided into the following three categories:

(1). Point outlier. In a time series, a data instance is a point outlier if its behavior is significantly different from

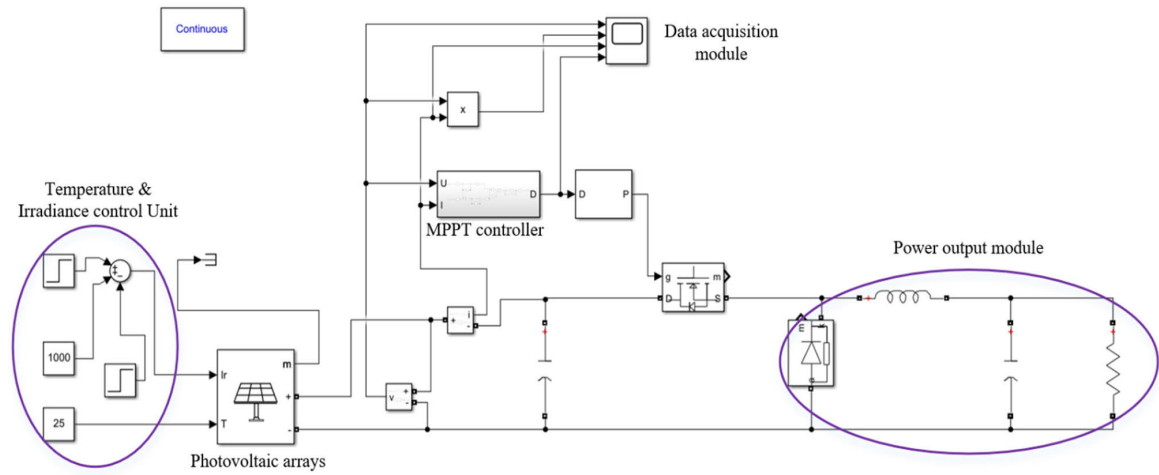


FIGURE 5. Simulink simulation of PV system.

other data [29]. Using Figure 6 as an example, the dashed lines represent the highest and lowest historical values of the behavior attribute in this data set. The blue circles represent normal points. The red crosses represent outliers. Data point a and data cluster A are typical point outliers.

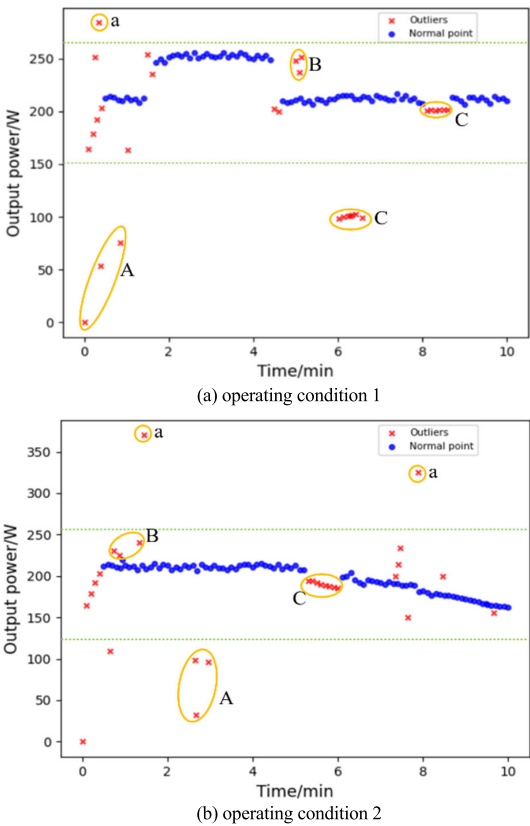


FIGURE 6. Outlier classification data graph.

(2). Situational outlier. In sequence data, a data instance is identified as a situational outlier (contextual outlier) if its behavioral attributes differ significantly from the rest of the

TABLE 2. Corresponding table of different operating conditions and simulation types.

operating condition	simulation type
operating condition 1	In cloudy weather, the movement of cloud layers causes changes in the output power of photovoltaic panels due to shading.
operating condition 2	The reduction of irradiance and temperature leads to the reduction of the output power of photovoltaic panels
operating condition 3	The ambient temperature increases (e.g. enhanced noise), resulting in an increase in the output power of photovoltaic panels
operating condition 4	In harsh environments, the output power of PV panels fluctuates sharply in a short period of time

data instances or from the overall trend of the data only in a particular scenario [30]. Such outliers can be abstracted into binary groups (behavioral attributes, situational attributes). The behavioral attributes are reflected by objective indicators such as temperature and irradiance, and describe the physical characteristics of the observed object [31]. Situational attributes are reflected by indicators that describe the background of data collection, such as time and location, and must to be analyzed in conjunction with behavioral attributes. As shown in Figure 6, time reflects situational attributes and output power reflects behavioral attributes. Each point in data cluster B behaves normally within its own range of behavior attributes, but deviates from the normal expectation at its own time point and time interval.

(3). Collective outlier. In time series data, collective outliers usually consist of a series of related data instances. When they appear together in a certain pattern, their overall performance deviates significantly from the normal expectations of the data, but each data instance does not constitute an outlier when analyzed separately [32]. In Figure 6, the observed range of neutron sequence C fluctuates within the normal range, but there is a significant difference between its corresponding time interval and the historical curve.

TABLE 3. Recall rate results of various algorithms under different operating conditions.

operating condition	algorithms /Recall rate						
	IF	SVM	LOF	GMM	GMM-RF	GMM-NN	GMM-IF
operating condition 1	0.73	0.41	0.62	0.57	0.70	0.32	0.90
operating condition 2	0.57	0.51	0.43	0.50	0.53	0.40	0.82
operating condition 3	0.88	0.52	0.64	0.76	0.68	0.55	0.92
operating condition 4	0.63	0.58	0.52	0.58	0.79	0.52	0.85

TABLE 4. Accuracy rate results of various algorithms under different operating conditions.

operating condition	algorithms /Accuracy rate						
	IF	SVM	LOF	GMM	GMM-RF	GMM-NN	GMM-IF
operating condition 1	0.87	0.81	0.87	0.88	0.85	0.74	0.95
operating condition 2	0.85	0.82	0.81	0.84	0.86	0.86	0.88
operating condition 3	0.91	0.83	0.89	0.91	0.88	0.86	0.93
operating condition 4	0.90	0.78	0.77	0.90	0.89	0.81	0.91

TABLE 5. F-value results of various algorithms under different operating conditions.

operating condition	algorithms /F-value						
	IF	SVM	LOF	GMM	GMM-RF	GMM-NN	GMM-IF
operating condition 1	0.74	0.52	0.72	0.71	0.71	0.39	0.88
operating condition 2	0.64	0.58	0.67	0.60	0.63	0.57	0.73
operating condition 3	0.80	0.55	0.71	0.77	0.69	0.62	0.86
operating condition 4	0.68	0.46	0.43	0.69	0.71	0.47	0.73

2) PHOTOVOLTAIC SYSTEM SIMULATION AND DATA ACQUISITION

Data collection of photovoltaic system is carried out based on the Simulink simulation model in Figure 5. The simulation model includes photovoltaic array, temperature and irradiance control unit, MPPT controller, data acquisition module and power output module. Among them, MPPT controller adopts disturbance observation method to track the output power of photovoltaic array. The data acquisition module is responsible for collecting MPPT output data and photovoltaic system operation data. Because the photovoltaic array in the operation process, the temperature, irradiance and other environmental factors changes and the occurrence of abnormal states, will lead to the photovoltaic array output power changes, so the use of different ways to simulate the change of environmental factors and abnormal states.

First of all, the temperature and irradiance control unit is used to simulate changes in the external environment. For example, a rise signal is connected to the input interface of the photovoltaic array to simulate the rise in the external

environment temperature or irradiance, so as to control the input of the photovoltaic array. The range of temperature is 25-60 °C, and the range of irradiance is 200-1200 W/m². Then, according to different types of outliers, each abnormal data type is simulated. For example, add data beyond the highest and lowest values of the behavior attribute to simulate point outliers. The proportion of abnormal data in the total data set is about 20%. According to this method, a total of 4 operating conditions were set up. In the first 3 operating conditions, 3 sets of simulation data were simulated by changing the irradiance and temperature change rate, and a total of 10 sets of simulation data were simulated. Each group contains 100 data points, totaling 1000 data points. Each data point includes five characteristics of photovoltaic array output power, current, voltage, temperature and irradiance. The sampling interval was 6 s. The total sampling duration is 100 minutes. The corresponding tables of different operating conditions and simulation types are shown in Table 2.

Outlier points were set respectively for 10 groups of data of the above four operating conditions. In the original data

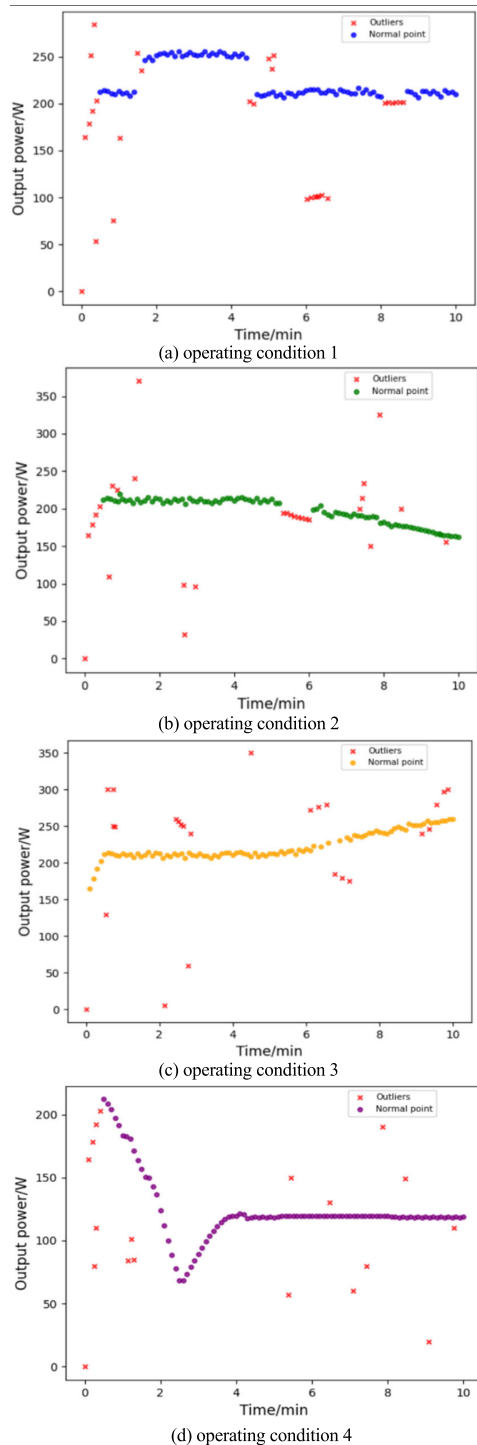


FIGURE 7. PV output power diagram under different operating conditions.

without outliers, there is a positive correlation between PV output power and irradiance on the whole. There is also a strong correlation between power and current. Temperature has little effect on power. The whole fault-free data can reflect the operation law of the system under normal working conditions. Part of the data set is shown in Figure 7. The colored circles represent normal data and the red crosses represent outliers.

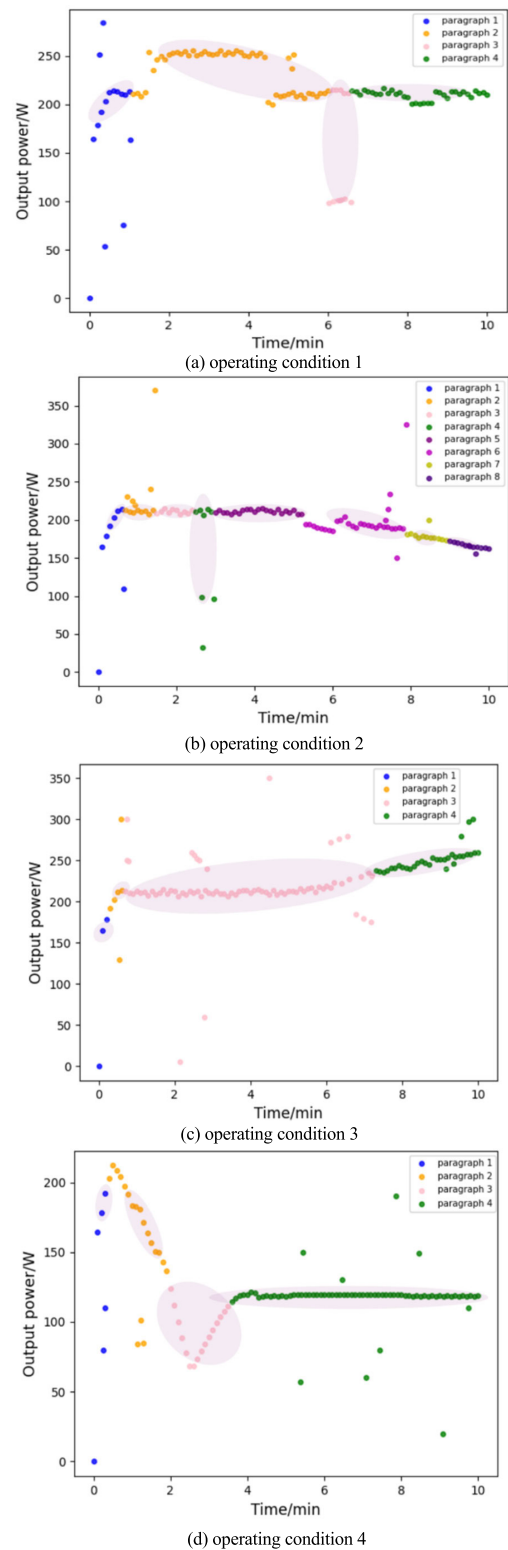


FIGURE 8. Segmentation and GMM fitting data trend charts.

B. SIMULATION EXPERIMENT AND COMPARISON

1) EFFECTIVENESS EXPERIMENT AND ANALYSIS

To verify the effectiveness of the algorithm. Firstly, the decision tree and related algorithms are used to process the

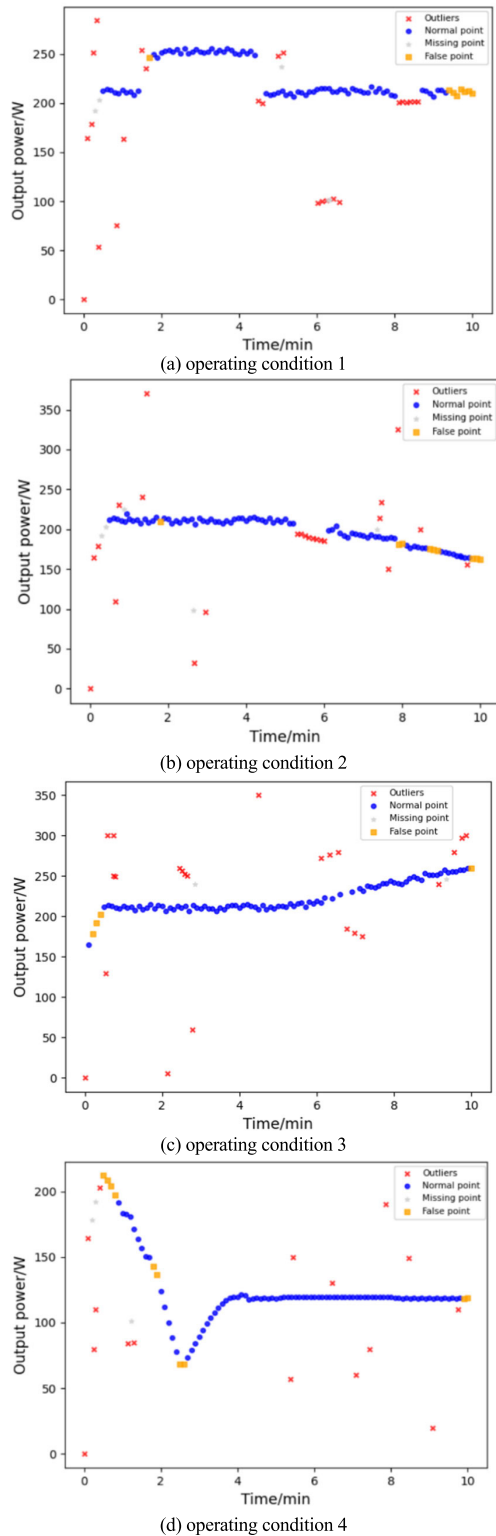


FIGURE 9. Detection results by GMM-IF algorithm under different operating conditions.

operation data in different operating conditions. Then, on the basis of each segment, GMM algorithm is used to fit the data. The fitting trend of the GMM algorithm is shown in

Figure 8. Figure (a)-(d) corresponds to operating conditions 1-4 respectively. Data points with different colors represent the data segmentation results after decision tree processing. The shaded part represents the Gaussian-like distribution trend of each segment after GMM fits the data sample.

From the experimental results, it can be seen that: under various working conditions, the decision tree algorithm can effectively segment data samples, and each shaded part can reasonably perform data fitting on each segmentation result. The method of using GMM for outlier detection after segmentation overcomes the premise assumptions of the algorithm to a certain extent.

Finally, on the basis of each segment, the coarse detection results are obtained by setting the threshold, and the IF algorithm is further used for joint decision-making judgment to obtain the final detection results. The final test results of the GMM-IF algorithm are shown in Figure 9, where the test results from a to d correspond to operating conditions 1-4. In the figure, blue circles represent normal points, red crosses represent outlier points, orange squares represent false detection data and gray pentagons represent missed data. In working conditions 1 and 3 with low nonlinearity, the number of false and missed detection data is significantly smaller. As the degree of nonlinearity increases, false and missed detection data increase, mainly concentrated around the segmentation results.

2) COMPARATIVE EXPERIMENT AND ANALYSIS

Firstly, the outlier detection results of each comparison algorithm are shown in Figure 10, taking condition 4 as an example. From a to f, the algorithms correspond to IF, SVM, LOF, GMM, GMM-RF and GMM-NN respectively. Each graph is annotated in the same way as Figure 9.

Then, the recall rate, accuracy and F-value are used as evaluation indexes respectively to quantitatively compare the GMM-IF with other algorithms to highlight the differences of each algorithm. The abnormal data is set as a positive class and calculated according to the following formula:

$$Recall = \frac{TP}{TP + FN} \quad (20)$$

$$F = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (21)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (22)$$

where, TP indicates that it is abnormal data and is detected as abnormal data. FN is abnormal data, but the detection is normal data. TN indicates that the data is normal and is detected as normal. FP indicates normal data, but abnormal data is detected.

The recall rate, accuracy and F-value of different algorithms under different operating conditions are listed in Table 3-5. Figure 11 shows the recall rate curves of different algorithms under different operating conditions. Figure 12 shows the accuracy curves of different algorithms under different operating conditions. Figure 13 shows the F-value

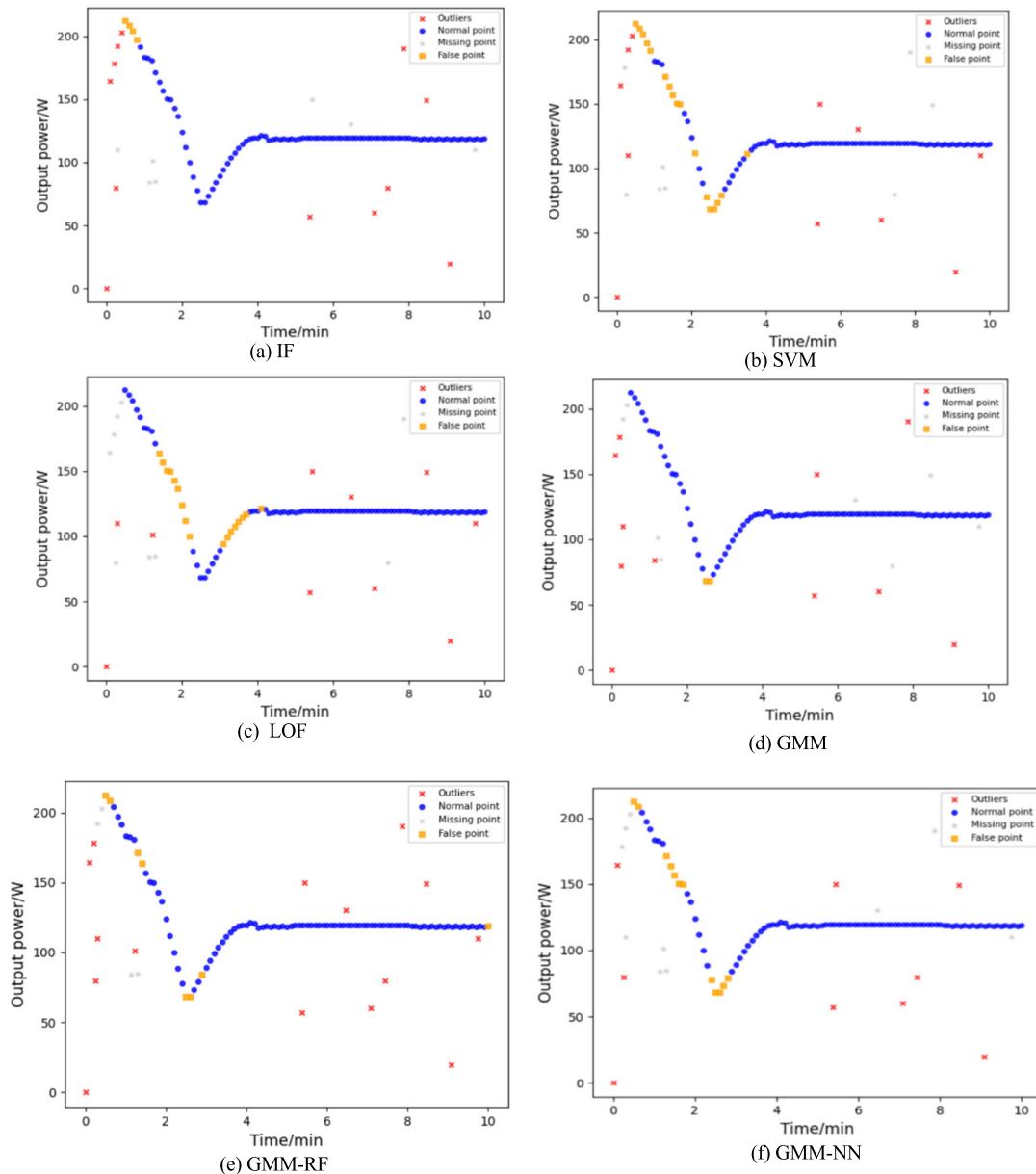


FIGURE 10. Outlier detection results of each comparison algorithm under operating condition 4.

curves of different algorithms under different operating conditions. The horizontal coordinate of Figure 11-13 represents different algorithms. The vertical coordinate represents recall rate, accuracy and F-value respectively. The curves with different colors represent different operating conditions.

According to the test results, the detection accuracy of the GMM-IF algorithm is the highest in the operating conditions with low nonlinear degree. For example, both the recall rate and accuracy reached more than 90% in the operating condition 1 and 3. In the operating condition 1, GMM-NN performed the worst. In the operating condition 3, the accuracy of GMM-NN and SVM is the lowest, the recall rate is close, the recall rate of GMM-RF is better than LOF, but

the accuracy is lower. The recall rate of GMM-RF is better than LOF, but the accuracy is lower. With the increase of the degree of nonlinearity, the performance of GMM-IF is still the best. The recall rate of SVM is getting higher and higher, but its accuracy is decreasing gradually. The LOF decreases significantly in both aspects.

In terms of F-value: In operating conditions 1 and 3 with low nonlinear degree, the GMM-IF algorithm performs the best, while the performance of GMM-NN and SVM is poor. In operating condition 1, the performance difference among the other algorithms is relatively small, but in operating condition 3, the IF and GMM algorithms outperform the other algorithms. As the nonlinear degree increases, the GMM-IF

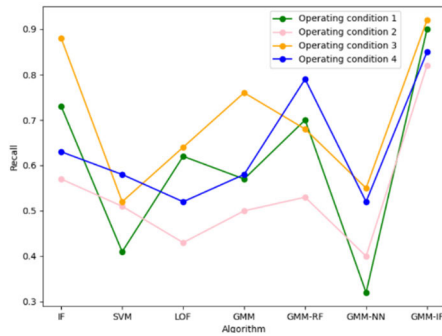


FIGURE 11. The recall rate curves of each algorithm under different operating conditions.

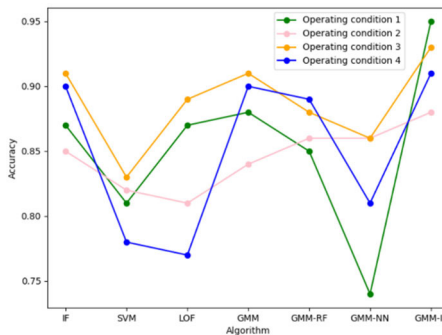


FIGURE 12. Accuracy curves of each algorithm under different operating conditions.

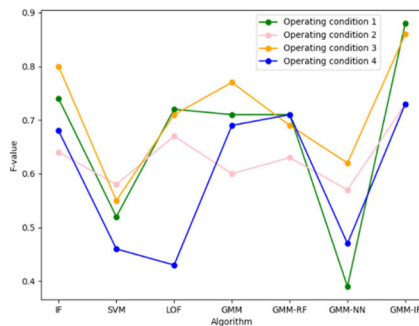


FIGURE 13. F-value curves of each algorithm under different operating conditions.

still outperforms the relevant comparison algorithms, while the performance of GMM-NN and SVM remains suboptimal. Additionally, the performance of the LOF algorithm gradually decreases, and the performance gap between GMM-RF and GMM-NN algorithms becomes minimal.

3) OVERALL RECOGNITION RATE

The classification detection accuracy of the GMM-IF algorithm for different outlier categories under different operating conditions of the overall data set is shown in Figure 14. The detection accuracy of different outliers in different conditions is different, which is caused by the combination of the distribution difference of simulated anomaly data and the non-linearity of the original data. However, the GMM-IF

algorithm still has a good performance: the detection accuracy of point outliers is the highest, followed by the detection accuracy of situational outliers. The collective outliers are mostly distributed near the normal sequence data, so the performance is poor.

C. EXPERIMENT AND ANALYSIS OF ACTUAL OPERATION DATA

1) PV ARRAY OPERATION DATA

Figure 15 shows the actual operation data of PV array in a county of Weifang City, Shandong Province. This data set includes only two characteristics: time and photovoltaic output power. The horizontal axis represents the time and data is collected every 10 minutes. The vertical axis represents the output power of the PV array in MW. Using pink circles for normal data and red crosses for outliers. The outlier points come from the comparison between manual data recording and data recording of intelligent acquisition equipment.

The PV array is laid on one side of monocrystal silicon, the maximum output power of each PV panel is 455 MW. A total of 13,182 pieces are assembled. The installation inclination is tiled with the roof. There are facing south of -5° and 5° . There was light rain and the temperature was $3-10^\circ\text{C}$. The field PV array is shown in Figure 16.

2) EXPERIMENT AND RESULT ANALYSIS

Before performing anomaly detection, data preprocessing is conducted first, including duplicate and missing data detection, duplicate data removal and missing data filling (using linear interpolation method). After data preprocessing, the GMM-IF algorithm is applied to detect outliers in real photovoltaic array operation data. The data trend of each segment fitted by GMM algorithm is shown in Figure 17. The final detection results are shown in Figure 18. The results show that the GMM-IF algorithm has a good performance in overcoming prior assumptions under the field operation data.

TABLE 6. Evaluation table of different algorithms under actual running data.

algorithms	evaluation indicators		
	Recall	Accuracy	F-value
IF	0.56	0.82	0.51
SVM	0.43	0.85	0.50
LOF	0.25	0.79	0.30
GMM	0.63	0.84	0.65
GMM-RF	0.31	0.77	0.32
GMM-NN	0.31	0.80	0.34
GMM-IF	0.84	0.88	0.65

Recall rate, accuracy rate and F-value are still used as evaluation indicators to evaluate the performance of different algorithms under the actual operation data of the photovoltaic array, as shown in Table 6. Plot it as a line graph, as shown in Figure 19. It is not difficult to see that under the actual photovoltaic power station operation data, the GMM-IF algorithm still has the highest detection accuracy, with a recall rate of 84% and an accuracy rate of 88%. The next best-performing



FIGURE 14. Detection rate of outliers of confusion matrix under different operating conditions.

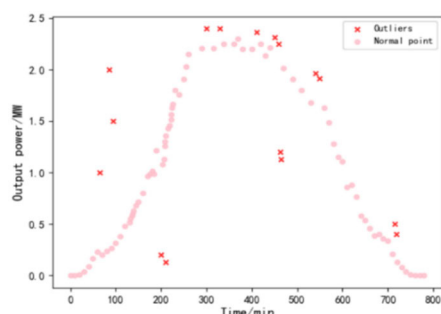


FIGURE 15. The actual operation diagram of a PV.



FIGURE 16. Field PV array.

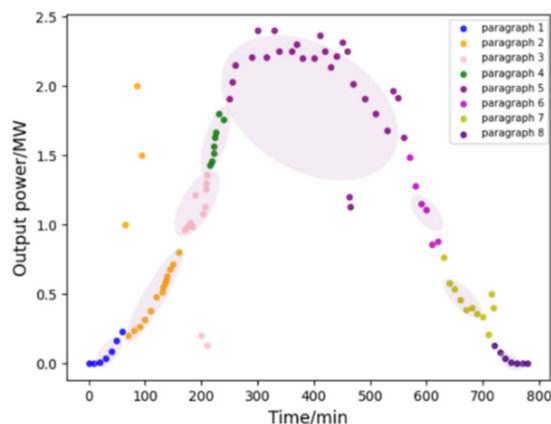


FIGURE 17. Segmentation and GMM fitting data trend charts.

algorithm is GMM, followed by IF and SVM, each with its own advantages. The IF algorithm achieves relatively high

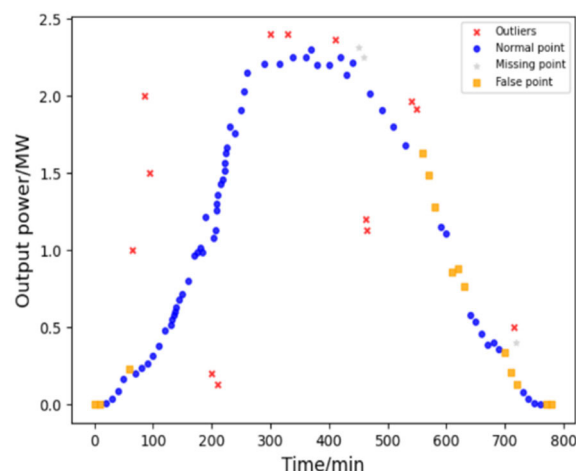


FIGURE 18. The final detection result of GMM-IF algorithm.

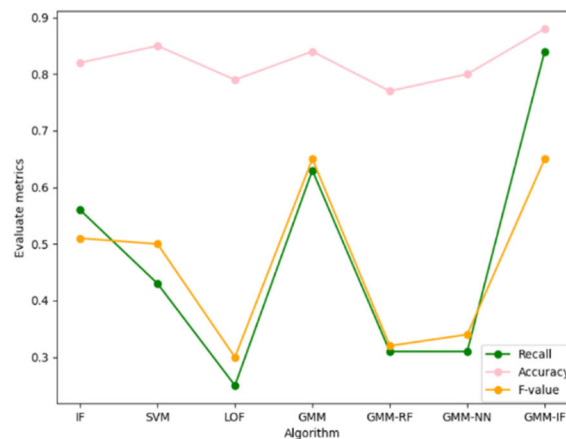


FIGURE 19. Curves of different evaluation indicators.

recall and accuracy but has a lower F-score, while SVM exhibits the opposite trend. The remaining three algorithms show lower detection accuracy with minimal differences between them.

V. CONCLUSION

Aiming at the problems of poor adaptability of traditional GMM algorithm and insufficient edge outlier detection ability under the background of high nonlinear degree,

a segmented outlier detection method based on GMM-IF is proposed. This method integrates decision tree segmentation, cross-validation, grid search, and the IF algorithm to improve upon the traditional GMM algorithm by overcoming its prior assumptions and enhancing detection accuracy. Compared with relevant comparison algorithms under each operating condition and actual photovoltaic array operating data. Quantitative calculation and qualitative analysis were carried out from three aspects of recall rate, accuracy rate and F-value. The following conclusions were drawn:

(1). The GMM-IF algorithm achieves high detection accuracy. By adopting a multi-model fusion strategy and using the isolation forest as a decision-making mechanism, the GMM outlier detection results are refined, effectively reducing the false positive rate and improving the detection accuracy of the GMM algorithm.

(2). Overcoming the prior assumptions of the traditional GMM algorithm. Compared with the traditional GMM algorithm, the method of segmental processing of PV data using decision tree and parameter optimization combined with cross-validation and grid search can be simpler and effective in time series anomaly detection.

(3). The GMM-IF algorithm has high practicality. Under different operating conditions, the GMM-IF algorithm achieves the highest recall, accuracy, and F-value, outperforming other comparison algorithms. It is also capable of detecting edge outliers that are close to the main output power of the photovoltaic array. It is suitable for photovoltaic system operation data with high nonlinear degree and edge outliers.

Although the GMM-IF algorithm demonstrates high detection accuracy in photovoltaic system anomaly detection and effectively overcomes the prior assumptions of the traditional GMM, it still has certain limitations. Firstly, the model is sensitive to parameter selection. Although optimization methods are used, different datasets and operating conditions may require innovative parameter tuning. Secondly, although the use of isolation forest to optimize the decision-making mechanism reduces the false positive rate, misjudgments may still occur under certain complex operating conditions. Finally, the algorithm errors come from the decision tree segmentation-related errors and joint decision-making defects. To address the above issues, future work will focus on the following two aspects: First, enhancing the model's adaptability, allowing the algorithm to automatically adjust in different scenarios and reduce human intervention. Second, further reducing the false positive rate by incorporating temporal features and multimodal data, strengthening the robustness of decision-making and improving the model's stability in complex environments.

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